

Exploiting Hierarchical Models for Street View Image Geolocalization

github.com/MatteoCamillo-code/GeoLoc-CVCS

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Abstract

Global image geolocalization is challenged by immense visual diversity and environmental ambiguity. Popular datasets, such as YFCC100M, often need intense preprocessing due to the presence of inherently non-localizable images (indoor, food, animals), but also containing many landmarks. On the other hand Street View image datasets need close to no preprocessing since all images are taken outside but contain less diversity and fewer clear landmarks (many similar roads with few significant cues about the location). In this second setting, we explore the effectiveness of a hierarchical multi-head architecture. We further utilize Individual Scene Networks (ISNs) to specialize the model on urban and natural environments. Our work integrates several modifications, including an Inception-v4 backbone, a Convolutional Block Attention Module (CBAM), and a weighted inference strategy utilizing cell density centers. Experimental results on a 10-state US subset of the OpenStreetView5M dataset (OSV5M), demonstrate that while hierarchical spatial knowledge improves accuracy across Great Circle Distance (GCD), on the contrary, scene-specific features do not play a significant role into improving performances over Street View data.

1. Introduction

Image geolocalization is the task of predicting a picture’s coordinates based solely on its visual content. It is an inherently difficult task, as many images lack distinct landmarks or contain ambiguous cues such as varying lighting and camera settings. Although many attempts focus only on specific cities, global scale prediction remains a challenge due to the immense visual diversity of the earth and the uneven distribution of pictures. Possible applications can range from providing location information when GPS metadata is missing or unreliable, supporting applications like organizing and searching large photo collections, verifying the provenance of online images (e.g., misinformation investigations), and providing situational awareness for

emergency response and environmental monitoring.

Modern approaches can be divided into two main categories: retrieval based and classification based geolocalization. Due to the latest achievements in the Deep Learning field and the computational advantages of the Convolutional Neural Network architectures, classification based methods have seen a growing interest over the years. In this paradigm, geolocalization is treated as a classification problem by partitioning the map into discrete cells. However, it faces a trade off: fine partitioning can improve local accuracy but risk overfitting due to fewer training examples per class, while coarse partitioning can be more robust but lack precision. Our project addresses these issues by using a hierarchical model [7] to capture features at multiple spatial resolutions. By integrating scene classification, specifically distinguishing between natural and urban settings, we aim at reducing data diversity and enabling the model to learn more discriminative, environment specific features.

Our proposed hierarchical framework, utilizing multi head spatial partitioning and density center weighted inference, significantly outperforms the baseline by increasing the 100 km test accuracy from 6.88% to 33.94%. These results highlight the effectiveness of integrating multi scale spatial knowledge to resolve geographical ambiguity in non curated street-view datasets.

2. Related Work

One of the first significant attempts at CNN-based geolocalization is PlaNet [14], which has set the ground for several current developments. In PlaNet the partitioning of the earth is performed according to the S2 Geometry library and then a CNN is trained on the partitioning. Indeed, our baseline model follows the pipeline provided by PlaNet. To improve upon this, we adopt multi head hierarchical architectures as described by the work of Muller-Budack et al.[7] which simultaneously learn from several partitions at different scales. This allows the model to capture features at multiple spatial resolutions, addressing the trade off between coarse level robustness and fine level precision.

Furthermore, we utilize the Individual Scene Network

(ISN) structure, also introduced by Muller-Budack et al.[7], to improve accuracy by reducing visual diversity. While their work applied this to indoor, natural, and urban scenes, we focus our implementation specifically on the natural and urban categories due to the nature of our dataset. By training separate networks for these settings, the model should learn more scene-specific features.

Other works have tackled the problem by modifying the partitioning technique: CPlaNNet [10] proposed a variation of s2 which exploits combinatorial partitioning in order to obtain more fine grained set of geo-cells from a combination of coarse level partitions. More recent works have been defining semantically interpretable partitionings by exploiting the existing administrative hierarchy and boundaries (e.g. Country/Region/City) from the Nominatim library [12] or more sophisticated mixed systems which combine administrative hierarchy and OPTICS clustering over training metadata to obtain a Voronoi partitioning with more visually coherent geocells (PIGEON) [4]

On the architectural side, we explored the impact of the Convolutional Block Attention Module (CBAM) [15] and experimented the effects of changing the backbone CNN model from the ResNet50 to a more modern Inception v4 [11].

3. Dataset

For this project, we utilize OSV-mini-129k [8], a specialized subset of the larger OpenStreet View-5M (OSV-5M) dataset [1]. OSV-5M is a large scale, open access collection of over 5.1 million georeferenced street view images, covering 225 countries and territories. Due to limited computing resources, we opted for this significantly reduced version rather than the full image set. While the parent dataset provides global coverage, the subset focuses on a geographically constrained region consisting of ten states in the United States, covering roughly 3 million km²: Texas, Kansas, Oregon, Nevada, Oklahoma, Montana, Idaho, Nebraska, New Mexico, and Wyoming.

The OSV-mini-129k dataset comprises 129,018 Street View images. The data is split into a training pool of 119,008 images and a test set of 10,010 images. To perform model tuning and selection, we further divided the training pool into a training and validation set using an 85%-15% split.

To refine the data for our Individual Scene Network (ISN) approach, we performed specific filtering based on scene content. We utilized a ResNet50 model to identify and remove images classified as "indoor" from the training set since they represented less than 1% of our US based subset and represented mainly images of car interiors and noisy or corrupted images.

To prevent overfitting and improve generalization, we implemented the following augmentation techniques: Ran-

domResizedCrop (sampling an area covering 70% to 100% of the original image with an aspect ratio between 3/4 and 4/3), RandomHorizontalFlip (with 50% probability) and RandomRotation (with 20% probability, up to 10°). Images were resized to a uniform dimension of 224 × 224 pixels to ensure consistency with the model's input requirements.

To improve the robustness of our model in the second phase of the project, we expanded the training pool by incorporating relevant samples from the MediaEval Placing Task 2016 (MP-16) dataset [6], selecting only those located within our ten target US states.

To ensure good data quality and relevance to the geolocalization task, we used the scene classification scores from our initial ResNet50 model to filter the MP-16 data. We retained only those images classified as "urban" or "natural" with a confidence probability exceeding 0.7. This threshold was necessary because the MP-16 dataset contains a high volume of ambiguous or non localizable content, such as close up images of animals, food, or people which provide no useful visual clues for geographical estimation. By removing "indoor" scenes and low confidence subjects, we reduced noise in the expanded dataset. Consequently, our final augmented training pool consists of 177,224 images for the training and validation split, while the test set remains fixed at 10,010 images to maintain evaluation consistency.

4. Method

In this section, we detail the foundational methodology used for our geolocalization framework (2), moving from geographical discretization to the implementation of the Individual Scene Network (ISN) architecture.

4.1. Geographical Partitioning via S2 Geometry

We treat geolocalization as a classification problem by subdividing the geographical space into a set of non overlapping cells. We utilize the S2 geometry library [3] to project the earth's surface onto an enclosing cube, where each face serves as a root node for a quad tree structure. To generate our hierarchical partitions, we apply an adaptive subdivision process: a cell is recursively divided into four smaller child cells as long as it contains more than a predefined threshold of images, τ_{max} . After the subdivision, all cells containing zero images are removed to eliminate areas with no representative data.

To capture features at different scales, we define three specific maximum thresholds: $\tau_{max} = 50$ for the finest partitioning (c_1), $\tau_{max} = 100$ for the middle partitioning (c_2), and $\tau_{max} = 200$ for the coarsest partitioning (c_3). This approach results in three distinct sets of classes $\{|c_1| = 4742, |c_2| = 2509, |c_3| = 1337\}$ representing various spatial resolutions. This hierarchical partitioning (1) is essential because it allows the model to capture features at



Figure 1. Hierarchical geographical partitioning of the 10-state US study area using S2 geometry. The three configurations illustrate the adaptive subdivision process based on image density: (left) fine partitioning with $\tau_{max} = 50$, (center) medium partitioning with $\tau_{max} = 100$, and (right) coarse partitioning with $\tau_{max} = 200$.

different scales; while a coarse partitioning increases performance at a regional or state scale, a finer subdivision is necessary for higher precision at the city scale.

4.2. Baseline and Multi Partitioning Architecture

Our baseline architecture consists of a ResNet-50 backbone [5] pre-trained on ImageNet-v2 [9]. To adapt this for geolocation, we add a final fully connected (FC) layer where the number of output neurons corresponds to the number of classes in the finest partition. During the initial baseline phase, the weights of the ResNet backbone remain frozen, and only the final classifier is trained to map extracted features to the correct geographical cell.

To improve upon the baseline, we implement a multi partitioning variant where the features extracted by the shared backbone are fed into three separate classification heads, one for each spatial resolution. This allows the network to simultaneously learn geographical features at different scales, resulting in a more discriminative model that leverages global context to refine local predictions.

4.3. Individual Scene Networks (ISN)

The final stage of our foundational method is the implementation of Individual Scene Networks (ISN). Before an image is processed for geolocation, it is passed through a separate ResNet-50 model pre-trained on ImageNet and fine tuned on the Places365 dataset [16] to perform scene classification. The objective is to categorize the image as either "natural" or "urban". We exclude the "indoor" category as these images represent less than 1% of our dataset.

Following this classification, the image is routed to one of two specialized models: an urban-specific ISN or a natural-specific ISN. Each model is trained exclusively on its respective data. The rationale behind this split is to reduce the visual diversity within the training set, allowing each model to learn more specialized and "stronger" features specific to its environment, such as architectural patterns for urban areas or vegetation types and landforms for natural scenes, rather than forcing a single model to learn the global appearance of all environments simultaneously.

4.4. Loss Function and Inference

For all models, we employ the standard cross entropy. During inference for the validation set, the predicted location for a query image is determined by selecting the cell with the highest probability after applying a softmax function to the FC layer output.

To ensure optimal training, we implemented an early stopping mechanism based on classification accuracy (the percentage of correctly identified cells). It is important to note that loss, classification accuracy, and geographical accuracy (geo-accuracy) are not strictly coupled. While classification accuracy treats every incorrect cell as a total failure, geo-accuracy measures the physical distance between coordinates. We acknowledge that a standard cross entropy loss does not account for spatial proximity; for instance, misclassifying a cell for another within the same city is penalized as heavily as a continent scale error.

Ideally, a more sophisticated approach would involve a localized label smoothing technique that distributes probability only to geographically adjacent cells. However, due to the computational and implementation complexity of such a custom loss, we maintain the standard cross-entropy. We use traditional accuracy for early stopping to monitor and prevent overfitting, while relying on geo-accuracy as our primary metric to evaluate the practical real world performance of the model.

For the multi partitioning architecture, we exploit the hierarchical knowledge by calculating the product of probabilities across the three spatial resolutions. We select the cell in the finest partitioning (the one with the highest number of classes) that maximizes this hierarchical product.

4.5. Evaluation Metrics

To evaluate the performance, we utilize the Great Circle Distance (GCD) between the predicted location and the ground truth. We report the geolocation accuracy as the percentage of test images whose predicted location falls within a specific radius of the true coordinates, specifically at thresholds of 1 km, 5 km, 25 km, and 100 km. For these foundational models, the predicted location is defined as the geographical center of the predicted cell.

5. Experiments

In this section, we evaluate the performance of our geolocation framework on the validation set (1). We begin by establishing a baseline and then systematically introduce a series of modifications in order to improve the model's accuracy. Each change is evaluated based on its impact on the Great Circle Distance (GCD) thresholds.

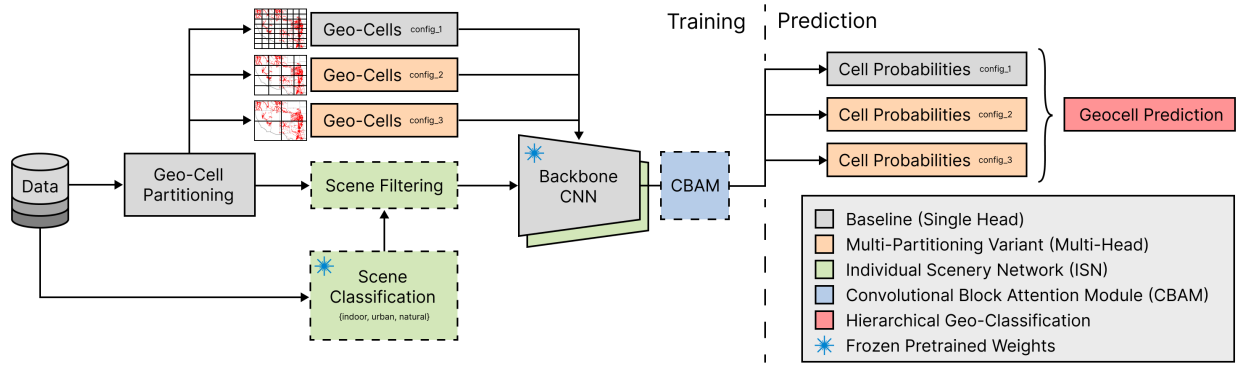


Figure 2. Overview of the proposed hierarchical geolocalization framework. The system integrates Geo-Cell partitioning across three scales, specialized Individual Scenery Networks (ISN) for urban and natural scenes, and a CBAM-enhanced backbone for refined feature extraction. Prediction is performed via a hierarchical multi-head strategy to produce the final geocell estimate.

5.1. Single Head Baseline Evaluation

Our initial experiment focused on a standalone ResNet-50 backbone trained exclusively on the finest partitioning level (c_1), following the classification concept introduced in the ISN paper. As shown in Table 1 (Conf. 1), the baseline achieved a 1 km accuracy of only 6.02%. While it demonstrates a baseline capability for regional localization (70.97% at 100 km), it struggles significantly with high visual intra class variance. These results indicate that high precision, city scale estimation is difficult to achieve with a single resolution classifier, as the model lacks the spatial context provided by coarser partitions.

5.2. Multi Head and ISN Comparisons

Following the initial baseline, we evaluated the Multi Head Hierarchical (Conf. 2) and the Individual Scene Network (Conf. 3) architectures. The Multi Head approach showed an improvement across all metrics, reaching 6.21% at the 1 km threshold. By learning across three spatial resolutions simultaneously, this structure allows the model to leverage “coarse to fine” spatial cues, where the coarser heads help narrow down the regional area while the finer heads pinpoint the specific cell.

In contrast, the ISN approach, despite being designed to reduce visual diversity, resulted in a performance decrease, particularly at the 1 km threshold (4.65%). We attribute this to the unique spatial distribution of our 10-state subset, which differs significantly from the global dataset used by the original authors. Specifically, we observed that the “Urban” partition performs worse than the “Natural” one because urban data is highly clustered within specific city centers, while natural data is more widespread across the territory.

This clustering makes the urban model more prone to high-magnitude errors; since city scapes often share similar

architectural and infrastructure characteristics across different states, the model can easily confuse a scene in Texas with one in Idaho. Conversely, such errors are less frequent within the “Natural” partition because the geographical areas sharing specific natural characteristics are comparatively smaller and more distinct in our subset.

Furthermore, unlike the global study where massive visual variance necessitates specialized models, our images are captured exclusively from a vehicular perspective and share a common environmental context. By splitting the dataset into two separate models, we likely hinder the network’s ability to perform *feature sharing*: in a unified model, geographical cues learned from “Natural” scenes (such as soil color or specific vegetation) can be effectively transferred to help classify “Urban” images that contain those same elements.

Within the ISN framework, the urban model is forced to focus exclusively on man made structures, thereby losing access to these subtle but critical regional indicators. While the complexity of a global dataset might justify such a split, the relative homogeneity of our vehicular, 10-state subset suggests that a single robust model is more efficient at capturing the interconnected visual cues necessary for accurate localization.

Based on these findings, we selected the Multi Head model as the primary foundation for all subsequent enhancements.

5.3. Ablation Study and Model Enhancements

To optimize the performance of the hierarchical and ISN models, we implemented and tested the following modifications iteratively.

5.3.1 Hierarchical Loss Weighting

To further refine the multi head architecture, we modified the training objective by transitioning from a simple mean loss to a weighted average loss. The weights w_i for each partitioning level i were defined as the inverse of the mean area of the cells within that partition, subsequently normalized to sum to one:

$$L_{total} = \sum_{i=1}^3 w_i \cdot L_{geo}^i$$

, where $w = \{0.61, 0.27, 0.12\}$. This strategy assigns the highest priority to the finest partition (c_1), which possesses the smallest average cell area. The rationale is to compel the model to prioritize high resolution learning while still utilizing the structural guidance and regional context provided by the coarser heads.

As shown in Table 1 (Conf. 4), although the metrics generally improved, reaching 6.34% at 1 km and 71.58% at 100 km, we did not observe a statistically significant breakthrough compared to the standard multi head model. We attribute this plateau to the inherent limitations of the cross entropy loss in a geographical context. As previously noted, while the weighted loss shifts priority, it still treats all misclassifications equally regardless of physical distance. We maintain that a more substantial improvement would likely require a selective label smoothing approach, where probability is distributed specifically to geographically adjacent cells, thereby penalizing "near misses" less severely than distant errors.

5.3.2 Hierarchical Inference and Density Centers

To maximize the predictive power of our multi resolution architecture, we adapted an inference formulation inspired by combinatorial partitioning logic. To reconcile scores across different resolutions, we utilize the hierarchical parent-child relationship: when a parent cell in a coarse partition receives a score, that probability is propagated to all its constituent children in the finest partition. We calculate the final score $s(g_k)$ for each fine-grained cell g_k as follows:

$$s(g_k) = \sum_{i=1}^N \frac{geo(g_k, C^i)}{\sum_{t=1}^K geo(g_t, C^i)}$$

where $geo(g_k, C^i)$ represents the softmax probability assigned by partition C^i .

Furthermore, we shifted from predicting the geometric center of a cell to predicting its density center, following the methodology proposed in the C-Planet paper. By selecting the top k cells (where $k = 1$ yielded the best results), we calculated the predicted coordinates l_{pred} using the mean

location of all training images e within those cells:

$$l_{pred} = \frac{\sum_{k \in \Gamma} \sum_{e \in g_k} geolocation(e)}{\sum_{k \in \Gamma} |g_k|}$$

To maintain precision, these calculations were performed using spherical coordinates before converting back to latitude and longitude. This method ensures the prediction is biased toward specific clusters of capture points, typically concentrated along road networks and urban centers, rather than an uninhabited mathematical center.

As reported in Table 1 (Conf. 5), this refined inference strategy resulted in the most substantial performance gain across all metrics. The 1 km accuracy increased from 6.34% to 13.45%, while the 5 km accuracy improved by over 11 percentage points. These results confirm that utilizing density centers effectively aligns the model's predictions with the actual underlying spatial distribution of the OSV-mini dataset.

5.3.3 Convolutional Block Attention Module (CBAM)

To enhance the network's representational power, we integrated a Convolutional Block Attention Module (CBAM) between the ResNet feature extractor and the fully connected classification heads. CBAM is a lightweight attention module that decomposes the attention process into two sequential steps: a *Channel Attention* submodule, which identifies "what" feature maps contain the most meaningful geographical information via global average and max pooling, and a *Spatial Attention* submodule, which focuses on "where" informative regions, such as landmarks or road boundaries, are located within the image.

The goal of this sequential refinement was to allow the network to "focus" on salient geographical features while ignoring irrelevant environmental noise like moving vehicles, weather artifacts, or glare. However, as reported in Table 1 (Conf. 6), the experimental results did not support this architectural addition. The 1 km accuracy dropped to 11.71%, and the 100 km regional accuracy decreased to 70.26% compared to the previous best configuration (Conf. 5). We observed that the attention maps may have over refined the features in a way that hindered generalization across the high visual diversity of our 10-state dataset. Consequently, due to this consistent performance degradation across all distance thresholds, we decided to exclude the CBAM module from our subsequent model optimizations.

5.3.4 Dataset Expansion (MP-16)

Observing that the initial ISN performed significantly better on "natural" scenes than "urban" ones, we hypothesized that the car-centric nature of the OSV-mini dataset limited the model's exposure to diverse urban environments. To

improve robustness, we augmented the training set with filtered images from the MP-16 dataset [13], specifically targeting urban and natural scenes within our 10-state region with a confidence threshold over 0.7. In this way, after adding the new images, the classes are now $|c_1| = 7702$, $|c_2| = 4042$, and $|c_3| = 2188$ (3). Despite the additional training data, we did not observe a clear overall improvement: the best model performs better at short distances but degrades at longer ranges. We attribute this trade-off to the increased complexity and visual diversity introduced by the augmented dataset. This dataset modification also affected the validation split, making results not directly comparable to models trained only on OSV-mini.



Figure 3. Hierarchical geographical partitioning of the 10-state US study area using S2 geometry using both the osv-mini dataset and the MP-16 dataset. The three configurations illustrate the adaptive subdivision process based on image density: (left) fine partitioning with $\tau_{max} = 50$, (center) medium partitioning with $\tau_{max} = 100$, and (right) coarse partitioning with $\tau_{max} = 200$.

5.3.5 Inception-v4 Backbone

Finally, we transitioned the backbone from ResNet-50 to a pre-trained Inception-v4 model. Inception-v4 utilizes a highly optimized architecture with inception residual blocks that allow for more efficient multi scale feature extraction. Given that geolocalization relies on cues ranging from tiny street signs to massive mountain ranges, the expanded receptive field and varying filter sizes of Inception-v4 were expected to provide a superior foundation for extracting discriminative geographical features. Moreover, Inception-v4 is typically trained with 299×299 inputs, which we also adopt to match the pre-training setup and preserve the expected feature statistics. The higher resolution helps retain finer details that the model can exploit.

However, as shown in Table 1 (Conf. 10), the integration of the Inception-v4 backbone using the parameters optimized in Configuration 7 led to a significant performance decline, with 1 km accuracy dropping to 5.66% and 100 km accuracy falling to 34.94%. Before this final transition, we performed a comparative evaluation between the Multi-Head and ISN architectures on the merged dataset (OSV-mini and MP-16), which confirmed the Multi-Head approach as the superior framework for this data distribution. Despite this structural selection, the results indicate that for our specific regional task, the Inception-v4 architecture did not provide the expected improvements and was outperformed by our previous ResNet based configurations.

5.4. Results on validation set

The results obtained by the evaluated models on the validation set are reported in the following table (1):

Table 1. Geo-Accuracy (%) on the validation set. In the following table, *Conf.* refers to the specific model configuration index. For cumulative setups, each row utilizes the optimized hyperparameters and training settings (*params*) derived from the referenced configuration. The bold models are the selected ones that will evaluate the test set.

Conf.	Model Description	1km	5km	25km	100km
1	Baseline (ResNet-50)	6.02	32.15	61.31	70.97
2	Multi Head Hierarch.	6.21	32.71	61.68	71.21
3	ISN (Urban/Natural Split)	4.65	29.87	60.89	71.12
4	Conf. 2 + Weighted Loss	6.34	32.31	61.80	71.58
5	Conf. 4 + Weighted Distance Inf.	13.45	43.85	65.23	73.00
6	Conf. 5 + CBAM Attention	11.71	39.36	61.21	70.26
7	Conf. 5 + MP-16 Dataset	13.91	38.19	54.15	61.03
8	Conf. 7 + CBAM	13.74	37.92	54.36	61.32
9	ISN + Conf. 7 params.	10.67	32.13	49.92	57.50
10	Inception-v4 + Conf. 7 params.	5.66	16.51	27.54	34.94

6. Results on test set

Here are reported the results of some selected models on the test (2).

Table 2. Comparative performance on the test set across different distance thresholds (km). Accuracy is reported as a percentage.

Conf.	Model Description	1km	5km	25km	100km
1	Baseline (ResNet-50)	0.01	0.15	1.60	6.88
2	Multi-Head + Weighted Inference	0.14	2.97	15.02	30.99
3	Multi-Head + Weighted Inf. + MP-16	0.15	3.07	16.47	33.35
4	Conf. 3 + CBAM	0.16	3.11	16.55	33.94

To evaluate the generalization capabilities of our framework, we compared the baseline ResNet-50 (Conf. 1) against our top performing configurations derived from the OSV-mini and the augmented OSV+MP-16 datasets. As shown in Table (2), while the proposed enhancements (Conf. 2-4) lead to a substantial improvement over the baseline—increasing the 100 km accuracy from 6.88% to nearly 34%—the absolute scores remain lower than those observed during validation.

We hypothesize that this performance gap stems from the test set’s lack of representativeness relative to the training distribution. The inherent “noise” and lack of curation in the OSV-mini dataset play a significant role; geolocalization is an exceptionally complex task where even minor inconsistencies in ground truth labels or image quality can severely hinder learning. This observation aligns with other related works in the field [2], which consistently report diminished performance when operating on non curated, real world street view data. Ultimately, these results confirm that while structural enhancements like Weighted Inference and MP-16 augmentation provide clear benefits, the quality of

the underlying dataset remains a critical bottleneck for high precision geolocalization.

7. Conclusions

This project developed a hierarchical learning framework for regional geolocalization using the OSV-mini dataset. Our results demonstrate that while a standard ResNet-50 baseline struggles with the high visual intra class variance of the 10-state study area, the integration of multi scale spatial knowledge and density-center weighted inference provides a robust solution to geographical ambiguity. The final proposed framework achieved a substantial performance leap, increasing the 100 km test accuracy from 6.88% to 33.94%.

Through a systematic ablation study, we identified that the Weighted Distance Inference utilizing cell density centers was the primary driver of high precision gains. Conversely, complex architectural modifications, such as the transition to an Inception-v4 backbone or the integration of CBAM attention modules, resulted in performance degradation or failed to yield improvements over less complex configurations.

Ultimately, this work highlights that high precision geolocalization remains heavily dependent on dataset quality. While our hierarchical approach effectively leverages "coarse to fine" spatial cues, the inherent noise in non curated street view data remains a critical bottleneck. Future research should explore localized label smoothing to account for geographical proximity during training and investigate advanced data cleansing, dataset selection strategies to further improve model discriminability in diverse regional landscapes,

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